

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 2

CONSUMERS BEHAVIOUR AND ITS ANALYSIS

DEFINITION AND MEANING:

Economics is a science of choice. It helps consumer to choose the best which satisfies his/her wants among the bundle of goods. If we look at our resources these are limited against our wants. So the objective of the consumer to satisfy maximum wants with limited resources is known as consumer's behaviour. In simple words the behaviour of the consumer regarding the selection, purchase and consumption of goods and services for the maximum satisfaction of his wants is known as consumer's behaviour.

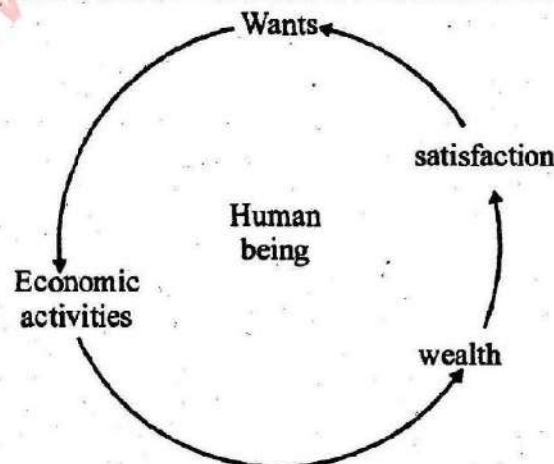
HUMAN WANTS AND THEIR SATISFACTION.

Human wants are also known as ends or necessities of life. From birth till death every body needs goods and services for his/her survival in this world e.g. food, cloth, shelter, upbringing of children, their education, health, transport, social contacts, security measures etc.

A large series of these wants is always hunting human beings. Human beings have to struggle and participate in economic and social activities for this purpose.

Initially these wants were simple and short-listed but with the improvement in science, technology, economic and social field these have become diversified, innovated and complex.

Following circle shows that human wants are the motives behind economic struggle.



Before going into detailed discussion regarding the laws, we need to elaborate some

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

basic concepts regarding human wants. Generally, we can divide goods into three broad categories with regard to consumption.

(i) Basic Necessities: The goods which are necessary for the survival such as; food, clothing, housing, medicines etc.

(ii) Comforts: The goods which facilitate human efforts in achieving satisfaction such as motorcycle, electric fan, watch, radio etc.....

(iii) Luxuries: The goods which add colours to our life or glorify the consumption style are called luxuries. e.g. want for T.V can be satisfied by 21" T.V. but to have 40" T.V is luxury. Similarly Air Conditioner and costly cars are also know as luxuries of life.

We generally divide wants into two broad categories.

(i) Economic Wants: Those wants for which we have to pay some price or have to participate in economic activity such as food, clothing, housing etc. These may be satisfied by goods or services.

(ii) Non-economic wants: Those wants which are fulfilled without any effort or economic activity are called non-economic wants, such as; oxygen, sunlight, rain, cool breeze, mother's affection etc. In economics we generally deal with economic wants.

FEATURES CHARACTERISTICS OF HUMAN WANTS

Following are the characteristics of human wants.

1. HUMAN WANTS ARE UNLIMITED:

As compared to the resources, human wants are unlimited hence these cannot be fulfilled simultaneously and require economic efforts to make them fulfill.

2. HUMAN WANTS ARE RECURRING :

Generally human wants recur after regular interval, for example the need to take meal, to wear cloths, to refill your car with petrol etc.

3. HUMAN WANTS DIFFER IN IMPORTANCE:

Some human wants are more important as compared to the other ones, so the important ones are needed to be fulfilled immediately whereas the less important can be postponed. For example, a hungry person will first take the meal rather to buy a pen or watch a movie.

4. ALTERNATIVE METHODS FEATURES OF UTILITY:

There are various ways by which a human want can be satisfied. For example, thirst can be quenched with water as well as Pepsis, Coke, Juices etc. Similarly for a long travel a person can use Rail, Car, Bus etc. Which ever is more economic and time saving.

5. COMPLEMENTARY WANTS:

Some goods are complementary if one such good is required for the satisfaction of human want the other one is automatically demanded, For example, pen and ink, car and petrol, CD and CD player. etc.

6. HUMAN WANTS COMPETE WITH EACH OTHER:

As resources are limited and wants are unlimited so, we try to fulfil those wants first which are relatively more important. Thus human wants compete with each other and the important wants drive out the less important wants.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

7. SOME WANTS BECOME HABIT:

Some wants are generated due to the customs, habits, fashion, conventions of the society. Such as smoking, reading books, magazine, listening music, etc.

8. QUALITY OF SATISFACTION:

Human wants can be satisfied with the availability of resources. Food can satisfy hunger, rail and bus facilities transportation etc..

2. GOODS AND SERVICES

Now we discuss the sources used to fulfill human wants. These are the goods and services which serve the purpose of satisfying human wants.

These can be classified as under.

(A) MATERIAL GOODS:

These are the goods which are tangible having weight and volume such as wheat, cloth, water, car, watch etc. These can be divided into (i) Free goods (ii) Economic goods.

(i) **Free goods or Non-economic goods:** These are the goods or services which are available to us without any effort or economic activity. For which we have not to pay any price for their use e.g rain, Air, sunlight, hot and cold weather etc.

(ii) **Economic goods:** Such goods or services are available to us at a certain price. We have to make some efforts or participate in economic activity for their attainment e.g. wheat, cloths, books, furniture etc.

(B) NON-MATERIAL GOODS OR SERVICES.

These are the goods which neither have volume nor are tangible. These are called the services which satisfy human wants-such as Doctor examines patient, Professor teaches to the student's, lawyer pleads his case etc.

(C) CONSUMPTION GOODS.

The use of the certain goods which directly satisfy human wants are known as consumption goods, for example, Water directly Quench human thirst, food satisfy hunger. So such goods directly serve the purpose for which they are used. These can be goods such as wheat, cloths, car, house etc. and services of Doctors, Professor, Lawyers etc. So consumption goods are comprised of both material and non-material goods (services).

(D) CAPITAL GOODS

Some goods do not directly satisfy human wants but these goods are used to produce certain other goods which satisfy human wants such goods are known as capital goods. The demand for these goods is known as "derived demand" whereas the demand for consumer goods is called direct demand. For example, House is a consumer good which directly satisfy the need of residence which is a "direct demand" but to construct house we need bricks, sand, cement, iron, wood, labour etc. which is derived demand for house. So the goods which are used to produce further wealth are known as capital goods. For example machines, tools, raw material, natural resources etc. Capital goods can be classified as under.

(i) **Raw Material:** For example raw cotton, sugar cane, leather, oil seeds etc.

(ii) **Intermediate goods:** For example: thread, wool, veneer sheets etc.

(iii) **Finished goods:** For example: machines, tools, tractors, tube wells, harvesters, wheat threshers etc.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Sometimes it becomes very difficult to classify a good as consumer good or capital good. Because If a sewing machine is used for personal use then it is called consumer good but if same machine is used by a tailor to sew clothes and earn money then it is called capital good. So according to Prof. Aistham the use of the good helps us in determining a good as capital or consumption good.

(E) PUBLIC GOODS VS PRIVATE GOODS.

In order to understand the concept of public and private goods, it is useful to group goods according to two characteristics.

- (i) Is the good excludable? Can the people be prevented from using the good?
- (ii) Is the good rival? Does one person's use of the good diminish another persons's enjoyment of it.

Keeping in view of the above two characteristics we can classify the goods as under.

(a) Private Goods: These goods are both excludable and rival. For example Ali's house or cloth or car is a private good. So these are the personal holdings and properties of someone. Use of Ali's car by someone else can reduce Ali's enjoyment of not using it.

(b) Public Goods: These are neither excludable nor rival. For example public parks, roads, bridges, national defence etc. So the use of these goods by one person do not prevent the others from using it. Similarly, the use of one person does not reduce the enjoyment of the other person.

(c) Common Resources: These are rival but not excludable. For example: fishing in the river is a rival good, when one person catches fish, there are fewer fish for another to catch. Yet these fish are not excludable good means it is almost impossible to charge fishermen for the fish that they catch.

(3) UTILITY

An early introduction of the utility notion into social sciences was accomplished by the English Philosopher Jeremy Bentham (1748-1831) After studying legal theory to draw some social legislation. He proposed that society should be organized on the "Principle of Utility" which he defined as "Property in any object to produce pleasure, good, or happiness or to prevent pain, evil or unhappiness". All legislations according to Bentham should be designed on utilitarian principles to promote "The greatest happiness of the greatest number".

The next step in the development of utility theory came when the Neo-classical economists such as William Stanely, Jevons (1835-1882) extended Bentham's utility concept to explain consumer behaviour. Jevons thought economic theory was a "calculus of pleasure and pains" and he showed that rational people would base their consumption decisions on the extra or marginal utility of each good.

So utility in Economics is taken as "quality of a good to satisfy certain human wants". Every thing or object in this world has its utility to satisfy human wants in various ways. For example, water has the utility to Quinch thirst, Bus has utility for transportation etc. a utility is a relative term. A good may have greater utility for one person but a little or no utility for the other. For example, pen has a greater utility for a student than for an illiterate person. So utility is relative with respect to time, space, use, demand, etc.

CHARACTERISTICS OF UTILITY

There are various factors upon which utility depends.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

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(1) Intensity of demand

Demand is important in determining the utility of a good or service. If the demand for a certain good is very intense, its utility will be high. For example, if a person is seriously ill, and he is in a dire need of certain medicine, that medicine will carry a high utility for that patient. But when he recovers the utility for that medicine will fall. Hence utility of a good or service is determined by the intensity of its demand.

(2) Use of good

Goods have various uses. The utility of a good or service in one use may differ from the utility of the same good in another use. For example, wood has various uses. Its utility when used in making furniture may be higher than its use as a coal or vice versa.

(3) Change in Form

When a good changes its form, its utility also changes accordingly. For example: when wood is taken from forest in blocks has different utility as when it is converted into useable items. Such as door, furniture. Similarly, clay has different utility, especially when converted into bricks.

(4) Changes in time and weather

Change of weather and time also change the utility derived from a good or service. Cold drinks and air conditioners have greater utility in summer as compared to in winters.

(5) Change in place

A good or service may have different utility at different places. For example, utility of a tractor increases more on agricultural lands than its use in cities.

(6) Change of ownership

Change of possession may also change the utility of good. If a computer is given to a person who is unaware how to use it, will yield no utility for him. Whereas a person who is fully aware of the operations will derive maximum utility from the use of computer.

(7) Change in technical knowledge

Change of technology due to scientific innovations and inventions also affect the utility of a good. For example a new technology may help a person to make the best use of a good when he knows all the features of a good.

(8) Utility and Benefit (usefulness)

Sometimes utility is taken as synonymous to benefits. It is wrong, because a good may have utility for a person but not beneficial for him. For example, wine may have utility for a drinker but it is not good for his health. Addictions and drugs may satisfy the need but these could not be beneficial in their effect.

Hence utility is a state of mind which is related with the satisfaction of certain needs or human wants. But the use of the good may or may not be beneficial morally, socially and ethnically.

UTILITY AND SCARCITY

In the process of learning economics one will be aware of the fundamental problem from which all economic problems stem that human beings have limited resources but unlimited wants. When wants exceed the resources available to satisfy them, that is "scarcity". Scarcity is everywhere. People want good health, long life, material comfort, physical and mental recreation and knowledge. None of these wants is completely satisfied

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

for everyone, and everyone has some wants that remain altogether unsatisfied. One can argue that everything is scarce but there is no shortage of anything.

So scarcity attached with utility makes economics a sciences of choice. Human being choose the best to maximize their utility. So scarcity create utility in different goods. If there would have no scarcity at all, economic problem could not arise. As all the human wants are satisfied then there will be no economic problem.

UTILITY AND INCOME

Poors have a little income as compared to that of Rich people. So their utility for income is high as compared to the rich people. This is why rich people spend lavishly upon goods having "demonstration effect". On the other hand poor people cannot afford it. They are very sensitive about their meagre income spending.

UTILITY AND PRICE

Utility and price have a close nexus. In analysis of consumer behaviour the equilibrium of the consumer is explained with the help of price and marginal utility. Consumer makes purchases upto the point where his marginal utility becomes equal to the price of the product. So price is used as a measuring rod for utility in consumer behaviour. If the demand for a product, especially in emergency, is very high. Then the utility of that good for consumer will also be very high and he will be willing to pay a high price for that product. So price and utility are helpful in explaining two important concepts Such as consumer's surplus and producers surplus which will be explained ahead. Later in this chapter.

DIFFERENT ASPECTS OF UTILITY

(i) Initial utility: Satisfaction achieved by the consumption of first unit of a good is known as initial utility.

(ii) Marginal utility: The change in the total utility of a commodity resulting from the use of it's additional unit is called Marginal utility. It can be calculated as $Mu = \frac{\Delta Tu}{\Delta q}$

(iii) Positive utility: A good is having positive utility when it's total utility is increasing and marginal utility has not become zero.

(iv) Zero utility: When point of satiety is reached by the consumption of good then total utility can not increase further or marginal utility becomes zero. This is known as zero utility.

(v) Negative utility: If a good is used after it's point of satiety then it's marginal utility becomes negative and total utility starts diminishing.

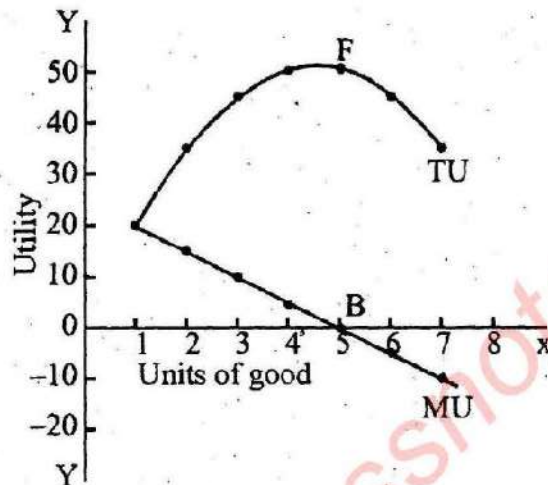
(vi) Total utility: It is the sum total of the utility derived from all the units of a good consumed.

All these concepts of utility are explained as under with the help of a schedule and diagram.

Units of good	Total utility (Tu)	Marginal utility (Mu)
1	20	20 → initial utility
2	35	15
3	45	10 Positive utility
4	50	5

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

5	50	0	Zero utility (point of satiety)
6	45	-5	Negative utility
7	35	-10	



RELATIONSHIP BETWEEN TOTAL UTILITY (TU) AND (MU) MARGINAL UTILITY

1. With the continuous use of the units of good Tu rises as long as Mu remains positive.
2. When $Mu = 0$ (at point B), Tu becomes maximum (at point F)
3. When Mu becomes negative, Tu start falling (after point F)
4. Right from the use of first additional unit Mu curve starts diminishing showing negative slope. While Tu curve initially rises, then remains the same and after that start falling.

MEASUREMENT OF UTILITY

Generally there are two schools of thought regarding the measurement of utility.

- (i) Cardinal measurement of utility, lead by marshall and his followers. They believed that satisfaction derived from the consumption of a good can be expressed in numbers like 1,2,3.....
- (ii) Ordinal measurement of utility, lead by R.G.D. Allan, J.R. Hicks. They were of the view that utility can not be measured in numbers or numerics rather it can be ordered according to preference.

Now we first explain cardinal measurement of utility approach with the help of following basic law of consumption.

LAW OF DIMINISHING MARGINAL UTILITY

This is the first law of consumption based on human psychology regarding use of a good. It is read as under in the words of marshall.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

“Other things remaining the same, marginal utility becomes smaller and smaller, the more of a commodity a person has.”

Mathematically $Mu = f(Q)$

OR “Other factors remaining constant, the marginal utility of a commodity diminishes with every increase in the stock of that commodity”

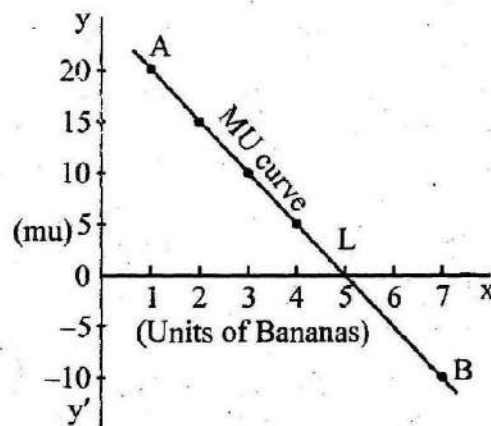
Now we explain this law. Suppose a person starts eating Bananas.

Units of Bananas	Total utility	Marginal utility $\frac{\Delta Tu}{\Delta Q}$
1	20	20
2	35	15
3	45	10
4	50	5
5	50	0
6	45	-5
7	35	-10

As shown in the table when consumer takes first unit of Banana his satisfaction is 20. Then he takes another unit. The second unit increases his total satisfaction by 35. So the marginal utility $Mu = \frac{\Delta T}{\Delta Q} = \frac{35-20}{2-1} = \frac{15}{1} = 15$. Hence the additional Banana has made the increment of 15 less than that of first unit which is 20. He continues to eat more bananas until he is fully satisfied at fourth unit where his satisfaction is maximum. Which is 50. Now if he takes the 5th unit it will bring no change in his satisfaction and still remained at 50 / Hence its Mu becomes zero. This is the point of satiety. If he takes one more unit his Tu falls from 50 to 45 making Mu negative i.e. -5.

It is further explained with the help of following diagram.

AB curve in the above diagram shows diminishing Mu curve. It intersects x-axis at point L where $Mu = 0$.



ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

ASSUMPTIONS OF THE LAW

Following are the assumptions which make the law workable.

(1) Continuously use of the good

If a good is not continuously used then this law will not hold true e.g. if a person is very thirsty and he is given one glass of water while the 2nd glass is given after a long gap then his utility will not diminish rather it may increase.

(2) Reasonable quantity

The commodity under use should be in reasonable unit. Neither too small nor too large e.g. if a person needs a full glass of water and he is given some drops of water his utility will increase rather decrease and will make the law unworkable. Similarly a tub of water cannot be assumed as one unit of water to drink.

(3) Homogeneous nature of good

The commodity under use should not differ in quality or characteristics otherwise utility will not diminish. For example, if first unit given to a thirsty person is a saltish glass of water and 2nd unit is purified with good taste then, his utility will not fall rather it may rise.

(4) State of mind of the consumer should not change

During the use of the certain commodity the mind set of the consumer should not change. For Example, if at 2nd unit of apple the consumer is reminded that doctor has advised him to take more units daily then his utility for apple will increase rather decrease.

(5) Goods should be divisible

Goods should be divided into small units so that these may be used accordingly.

(6) No change in consumer's income

Change in income, brings about changes in the purchasing power of the consumer which in turn changes the utility of the commodity used.

EXCEPTIONS / LIMITATIONS OF THE LAW

Following are the limitations where law is not applicable.

(i) Wealth and Money

As the wealth of a greedy person increases his utility or satisfaction also increases. Hence law of diminishing marginal utility does not apply on wealth accumulation.

(ii) Knowledge and Fine Arts

A person feel more satisfied as he acquires more knowledge. However continuous reading of books may diminish his utility but feeling of acquiring more knowledge gives him more satisfaction. Similarly learning of some art and attaining proficiency in it also increases its utility.

(3) Rare and most precious goods

Law of diminishing marginal utility also does not apply on acquiring more precious and rare goods such as diamonds or pearls. More a person have more he feels satisfied.

(4) Historical Findings

If a person is fond of collecting stampes, coins, utensils etc, the more he acquires, more he feels satisfied.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

(5) Narcotics and addictions

This law also does not apply on addictions such as wine, cigarette, heroin, drugs etc. the more an addicted person takes it, the more he feels satisfied and the greater urge he has.

(6) Ostentation / Fashion and show off

This law also does not apply on things which are in fashion or which are the symbol of prestige. Such as jewellery, costly cloths and certain brands of the product.

PRACTICAL IMPORTANCE

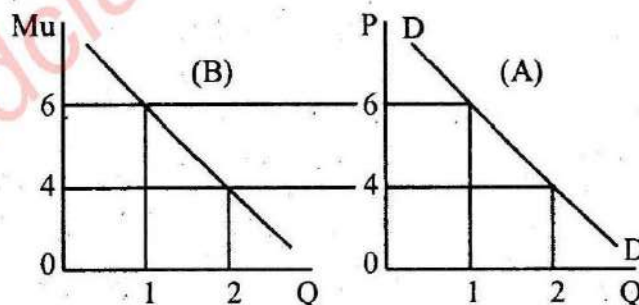
This law has great practical importance. It is used for explaining various aspects of economic theory.

(1) It guides consumer to maximize satisfaction.

When a consumer pays a certain price for a commodity, it can be taken for granted that he thinks that the satisfaction is at least equal to the price paid i.e. he continues to purchase that commodity until its falling Mu becomes equal to price. $Mu = p$. Hence we can say that price measures the marginal utility or that marginal utility indicates the price. So price and Mu move together. It is, however, wrong to say that the marginal utility governs or determine price, it simply indicates it. Instead both Mu and p are governed by demand and supply so. If $mu > p$ consumer will purchase more of good which will result in decreasing Mu until they become equal, $[Mu = P]$ If $mu < P$ consumer will decrease quantity so that Mu will increase and becomes equal to price $Mu = p$.

(2) Forms the basis of the law of demand.

As we said earlier consumer equates P & Mu for maximum satisfaction. As price and Mu move together. It forms the bases of law of demand. When price falls from Rs. 6 to Rs. 4 (A) consumer purchases more and hence its marginal utility falls from 6 to 4 (B).



(3) Forms the basis of theory and practice of Taxation

The value of money to a rich man is not so great as it is for poor. Hence the system of progressive taxation is used to levy higher taxes to higher income people and lower to lower income. So it also helps in making the distribution of wealth equal.

(4) Diamond – water paradox

Diamond, though not essential for life, command high price because of high marginal utility, which is due to scarcity. Water, though essential for life, is free or carry low price because it is abundant and have low Mu. As we know $Mu = f(Q)$ so thing which is abundant like water has low marginal utility and diamond which is scarce have high marginal utility and carry high price because $Mu = P$.

LAW OF EQUI-MARGINAL UTILITY

We all know that our wants are competitive. We have to make a choice between the more urgent and the less urgent wants. When we choose to buy a little more or a little less of

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

a commodity. We, in fact are trying to balance the marginal utility of the commodity and that of money. We in fact try to adjust the quantity of different goods, by changing our purchasing plan in such a way that the marginal utilities of all the goods purchased become equal. This law is applied when consumer purchases two or more goods. So it can be defined as

“Consumer is said to be in equilibrium when he spends his income on various goods in such a way that marginal utility of money to him in each direction of his purchase is the same”.

OR

“A consumer is said to be in equilibrium when he spends his money on various goods in such a way that Mu of every Rupee that he spends on each good is equal”.

“The law of equi-marginal utility states that maximum total utility will be achieved only if consumer divides his expenditure on various goods in such a way that marginal utility of the last unit of money spent on various goods remains the same”.

EXPLANATION

Suppose a consumer has 5 units of money which he wants to spend on two goods, flour and ghee. The marginal utilities of two goods are given in the following table.

Note: one unit of money = 100 Rs.

Money units	Mu of Flour	Mu of ghee
1	24	20
2	20	16
3	16	12
4	12	18
5	8	4
Total utility	80	60

If consumer spends first unit of money on flour it's marginal utility is 24. If 2nd unit of money is also spent on flour it's Mu falls to 20. Similarly, if all five units of money are spent on flour it's total utility derived is 80. Similarly if all the five units of money are spent on ghee its total utility becomes 60. But consumer wants to purchase both flour and ghee. So he will spend two units of money on Ghee and three units on flour. Because according to this arrangement the Mu of both goods is equal i.e. $16 = 16$ and consumer is spending his total budget. The satisfaction of the consumer is also maximum because

$$\text{Total utility from flour} = 24 + 20 + 16 = 60$$

$$\text{Total utility from Ghee} = 20 + 16 = 36$$

$$\text{Aggregate utility of both goods} = 96$$

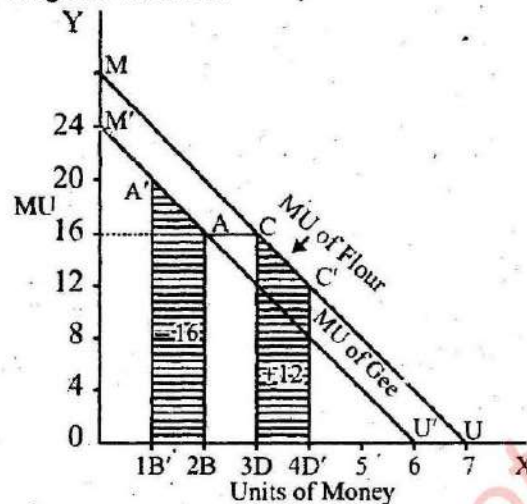
So the total utility from both goods is 96 which is higher than the total utility of flour which is 80 and total utility of Ghee which is 60.

Now if consumer would buy any combination where marginal utilities are not equal his total satisfaction would fall. For example, if consumer wants to spend one additional unit of money on flour i.e. $Tu \text{ of flour} = 24 + 20 + 16 + 12 = 72$.

Now consumer is left with one unit of money to spend on ghee where it's Mu is 20. thus total utility from both goods is now $72 + 20 = 92$ which is less than 96. So consumer could not maximize satisfaction if, he is not following the principle of equalizing marginal utilities of two goods

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

It is shown in the diagram as under:



It is clear from the diagram that when consumer was spending two units of money on Ghee and three units on flour his satisfaction was maximum. Because MU of both goods was equal $16 = 16$. Now if consumer spends one additional unit of money on flour i.e. 4th unit the addition in MU of flour is of 12. As he is left with one unit of money to spent on Ghee so decrease in MU of Ghee is $= 16$. As seen decrease (16) is greater than the addition (12) so there is a loss of 4 units.

In the previous example we have not used prices of the commodities to explain consumer's equilibrium. Now we explain by taking prices. As we have discussed earlier that consumer always equates the price of a good with its marginal utility i.e. $MU_x = P_x$.

Similarly consumer, if purchasing more than two goods will Equate Marginal utilities to their prices such as

$$MU_x = P_x$$

$$MU_y = P_y$$

$$MU_n = P_n$$

If we divide both sides by prices.

$$\frac{MU_x}{P_x} = 1$$

$$\frac{MU_y}{P_y} = 1$$

$$\frac{MU_z}{P_z} = 1$$

$$\text{So } 1 = 1 = 1$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Hence $\frac{Mu_x}{p_x} = \frac{mu_y}{p_y} = \dots\dots\dots = \frac{MU_z}{P_z}$

So consumer is said to be in Equilibrium when the ratio of the Mu and prices of all the goods become equal.

CONSUMER'S EQUILIBRIUM

Suppose consumer has Rs.20. Where as

$P_x = \text{Rs } 4 \text{ per unit}$

$P_y = \text{Rs } 3 \text{ per unit}$

Units purchased	Mu_x	Mu_y
1	34	30
2	32	28
3	30	26
4	28	24
5	26	22
6	24	20

According to the principal of consumer's equilibrium he will purchase two units of x where it's Mu is 32 and four units of y where it's Mu is 24 because at this combination.

$$\frac{Mu_x}{P_x} = \frac{Mu_y}{p_y}$$

$$\frac{32}{4} = \frac{24}{3}$$

$$8 = 8$$

At this point consumer is spending all his budget on two goods while deriving maximum utility.

Because

$$Y = Q_x \cdot P_x + Q_y \cdot P_y$$

$$20 = 2(4) + 4(3)$$

$$20 = 8 + 12$$

$$20 = 20$$

So by following this principle not only consumer is efficiently spending his total income i.e. 20 but also maximizing satisfaction i.e.

$$\begin{aligned} \text{Total utility of x} &= 34 + 32 = 66 \\ \text{Total utility of y} &= 30 + 28 + 26 + 24 = 108 \\ \text{Sum total of utilities of both goods} &= 174 \end{aligned}$$

Now suppose that price of x falls from Rs.4 to Rs.3. in order to maximize satisfaction consumer will change his purchasing plan by increasing the Quantity of x such that again

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

$$\begin{aligned}\frac{M_{ux}}{P_x} &= \frac{M_{uy}}{P_y} \\ \frac{24}{3} &= \frac{24}{3} \\ 8 &= 8\end{aligned}$$

Hence consumer will purchase six units of x because price of x has reduced and now again the price ratios of two goods are equal.

CONSUMER'S DIS-EQUILIBRIUM

When consumer ignores the principle of $\frac{M_{ux}}{P_x} = \frac{M_{uy}}{P_y} = \dots = \frac{M_{uz}}{P_z}$, he becomes into dis-equilibrium-position.

Keeping in view the previous example, if we assume that:-

- (i) Consumer purchases only one unit of x, thus $\frac{M_{ux}}{P_x} = \frac{34}{4} = 8.5$
- (ii) He purchases five units of y, thus $\frac{M_{uy}}{P_y} = \frac{22}{3} = 7.33$ As $8.5 \neq 7.33$

Hence Total utility of both commodities has decreased

Tu of x = 34

Tu of y = $30 + 28 + 26 + 24 + 22 = 130$

Tu of x + Tu of y = $34 + 130 = 164$

Which is less than equilibrium position.

Tu = 174

Shortage in Tu = $174 - 164 = 10$

It is due to dis-equilibrium-position.

ASSUMPTIONS

(1) Consumer is Rational

Consumer is supposed to spend his income rationally by purchasing more of two goods within his budget.

(2) Cardinal Measurement of utility

Utility can be measured in units or numbers.

(3) Diminishing marginal utility

The utility of every additional unit diminishes as consumer increases the amount of that good.

(4) Total utility is additive

The utility derived from the use of different goods is additive.

(5) Constant marginal utility of money

In fact the marginal utility of a commodity is measured in terms of money. So money is measuring rod of utility. If measuring rod is itself changing then it is not possible to

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

measure marginal utility. As a person has to part with money when make more purchases so his Mu of money must rise but it is assumed constant.

(6) Divisibility of Goods

Goods are divisible so that a consumer could use them according to his needs.

(7) Utilities are Independent

The utility derived by using one good is not affected by the utility of another good.

LIMITATIONS

(1) Cardinal measurement of utility is not possible

Utility is a state of mind whose numerical measurement is not possible. So it is an absurd idea.

(2) No Rational and Conscious Calculations

In fact most of our expenditure is governed by habits. There is no much of conscious calculation and careful weighting of the utilities.

(3) Ignorance of consumer

Consumer may not be aware of all the useful alternatives. So ignorance of consumer may hamper utility maximization.

(4) No fixed accounting period

The variation in the durability of different goods create problem regarding the maximization of satisfaction of a good in particular period.

(5) Indivisibility of certain goods

All goods are not divisible into small bits to enable consumer to equalize marginal utilities. Hence law remains only a theoretical concept.

(6) Effects of publicity

Effects of publicity and brand loyalty sometimes makes the law inoperative and consumer regardless of high price continues to purchase more of that good.

PRACTICAL IMPORTANCE

(1) Its Applications in Public Finance

It guides Finance Minister for levying taxes in such a way that the marginal sacrifice of each tax payer is equal. Moreover, the government makes use of this law while deciding about projects and their expenditures. Government tries to equate the social marginal utility of each project.

(2) Guidance for Consumer

In the pursue of utility maximization, this law guides consumer that he must substitute goods of greater utility for those possessing less utility, till marginal utilities are equalized.

(3) Guides in pricing the factors of productions

National income is the result of combined efforts of four factors of production i.e. land, labour, capital and organization. According to this law the reward of each factor is

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

determined regarding the marginal productivity. So factor with greater productivity would receive greater reward.

(4) Guidance of Producer

The objective of producer is to maximize profit which he can obtain by having the least cost combination. For this purpose he will be continuously substituting one factor for another till the marginal productivity of all the factors have been equalized.

ORDINAL APPROACH OR INDIFFERENCE CURVES APPROACH

The Economists like Prof. Edgeworth, Pareto, J.R. Hicks and R.G.D. Allen used indifference curve analysis to explain the utility maximization problem. They criticized cardinal approach as irrational and absurd. Because it is very difficult to explain human feelings in numbers so they used ordinal or indifference curves approach to explain human satisfactions. According to this approach consumer can better express his satisfaction by ranking his preferences rather giving numbers to them.

INDIFFERENCE CURVE

An indifference curve is a locus of points or combination of goods each of which yields the same level of satisfaction to which the consumer remains indifferent.

Or simply we can say.

“An indifference curve indicates various combinations of two goods which gives the consumer, the equal level of satisfaction.”

Now we explain it with the help of diagram. In fact indifference curve is derived with the help of law of diminishing marginal rate of substitution which is explained below. Marginal rate of Substitution (MRS) may be defined as the ratio between small quantities of the commodities which are equally valued by an individual.

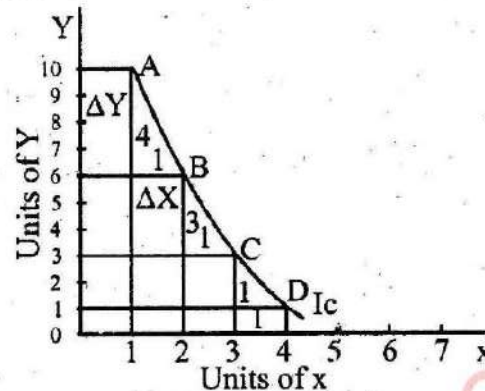
Suppose consumer wants to have two goods x and y and he has various combinations of two goods in his mind which give him equal level of satisfaction.

Combinations Of goods	Good x Units	Good y Units	MRS	Level of satisfaction
A 1 st	1	10	-	B
B 2 nd	2	6	1 : 4	B
C 3 rd	3	3	1 : 3	B
D 4 th	4	2	1 : 2	B

in the first combination consumer have one unit of good x and 10 units of good y. He wants to increase the quantity of x. In the second combination for one extra unit of x consumer is foregoing four units of y. It is because in order to keep the level of satisfaction of the consumer same on the same indifference curve when one good is increased i.e. x the other good i.e. y must be taken away from him. So when there is sense of gain alongwith sense of loss his satisfaction would remain same between the two combinations. Here MRS is 1:4 as

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

consumer moves to the third combination for one extra unit of x he is foregoing less of y i.e. MRS is 1 : 3 this is so because the utility for the good which consumer is losing increases successively as the quantity of y decreases but his satisfaction level remains the same, throughout the curve as under.



$$\begin{aligned} \text{Slope of I.C.} &= \frac{\text{Rise}}{\text{Run}} = -\frac{\Delta y}{\Delta x} = \frac{M_{ux}}{M_{uy}} \text{ because} \\ -\Delta y \cdot M_{uy} &= \Delta x \cdot M_{ux} \\ \frac{\Delta y}{\Delta x} &= \frac{M_{ux}}{M_{uy}} \end{aligned}$$

PROPERTIES OF INDIFFERENCE CURVE

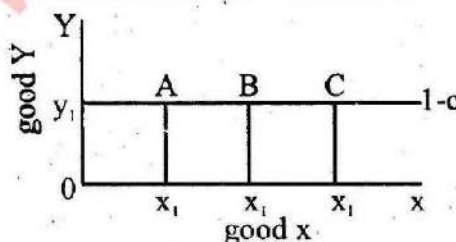
Following are the properties of Indifference curve.

- (i) An indifference curve slopes downward from left to right.
- (ii) Two indifference curves never intersect each other.
- (iii) Higher I.C. shows higher satisfaction and vice versa.
- (iv) Indifference curves are convex to the origin (0).
- (v) Indifference curve never touches any axis.
- (vi) Each indifference curve may not necessarily parallel to each other.

EXPLANATION

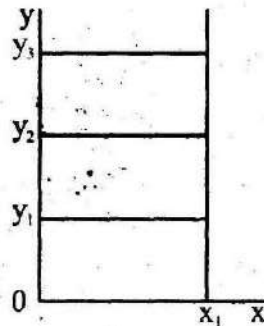
1. Indifference curves are negatively sloped from left to right.

It is because in order to keep the same level of satisfaction along the same I.C. when quantity of one good is increased the quantity of other is decreased. Imagine: following indifference curve can we keep satisfaction same at various combinations of the curve, if no then this cannot be an indifference curve.

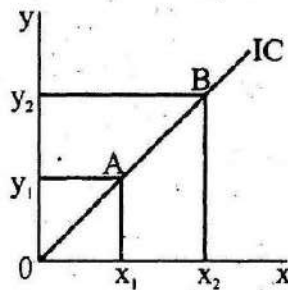


Here we are increasing x without losing y. Thus satisfaction will increase rather remaining same at points A, B and C

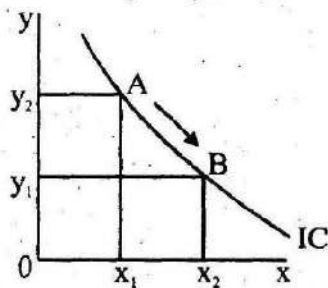
ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)



Similarly here quantity of y is increasing without losing x.



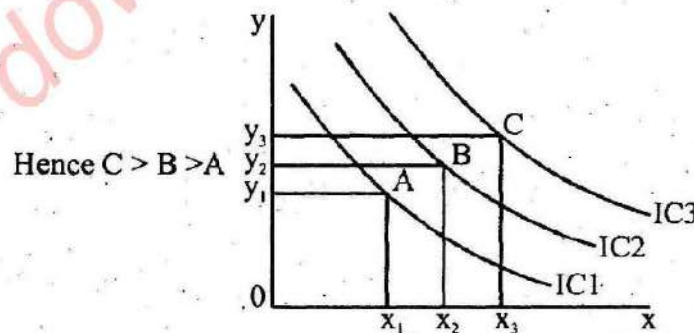
Here both x and y are increasing so point B is giving more satisfaction than point A.



This in reality is an indifference curve sloping downward from left to right.

2. Higher the I.C. higher would be the level of satisfaction.

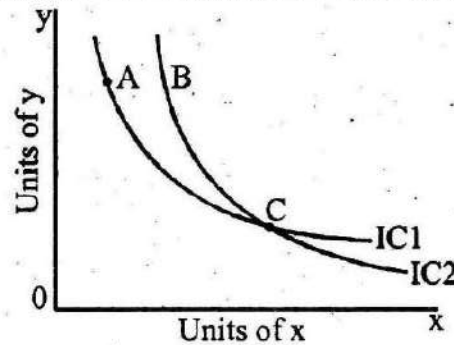
This means that on higher I.C. curve consumer is having more of both x, and y goods, so he will feel more satisfaction. At point C of IC₃ is great satisfaction than point B at IC₂ similarly, point-B gives more satisfaction than point A.



3. Two indifference curves never intersect each other.

It simply means that higher level of satisfaction can not be equal to lower level of satisfaction.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

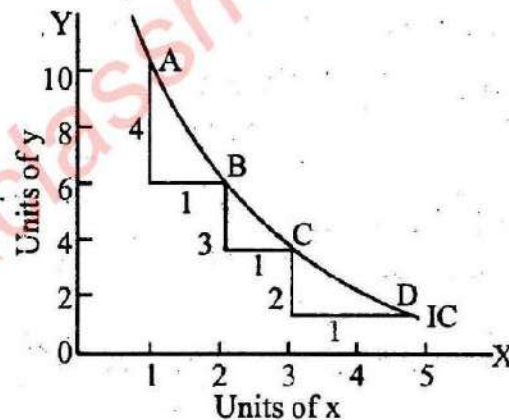


ON IC₁ A = C
 ON IC₂ B = C

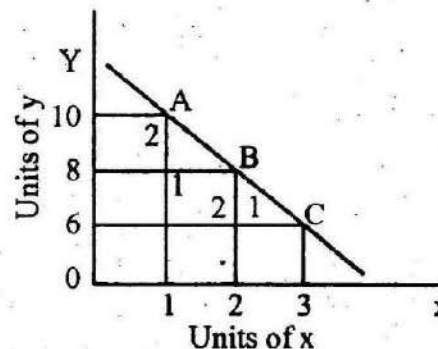
But it is not true because B is on higher I.C. than A. So $A \neq B$ rather $B > A$.

4. Indifference curves are convex to the origin.

The convexity of I.C. is due to diminishing marginal rate of substitution between two goods. It is because of the principle that $M_u = F(Q)$ which states as the quantity of one good increases its M_u falls and if, decreases its importance or satisfaction increases. Suppose: if rate of substitution is not falling then what will be the shape of IC.

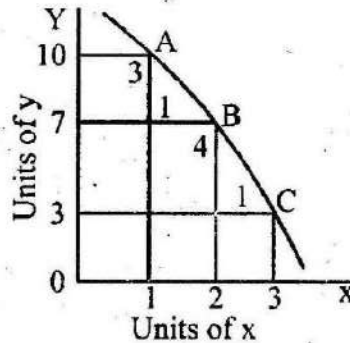


If I.C. is straight line then MRS between x and y is constant i.e. for one more x we are giving up two units of y at every combination.

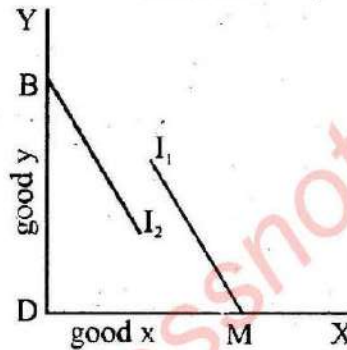


ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

If it is concave then MRS between x and y is increasing which is not practically possible.

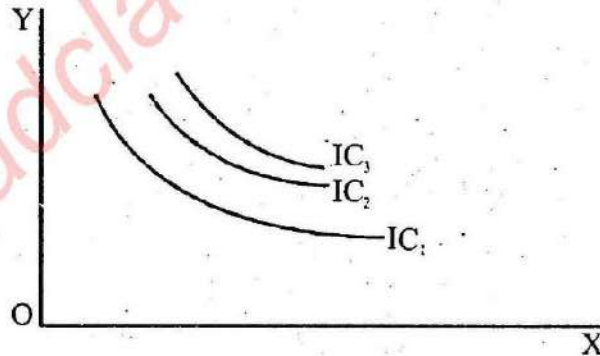


5. **I.C. never touches any axis**, if so, it means that consumer is satisfied completely by having only one good either x or y. this is against the hypothesis that consumer will be satisfied more by having more of both x and y. It may happen but in especial cases where consumer has no other choice and the other good is less preferred.



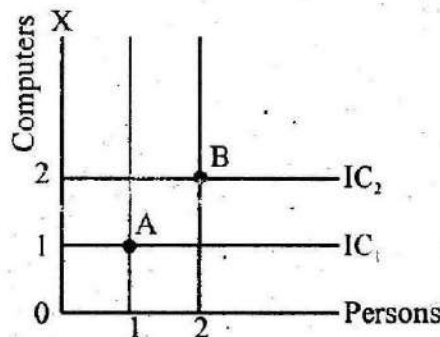
6. **Indifference curves may not parallel to each other**

Although I.C. are convex to the origin, however they may not necessarily be parallel to each other. It means that the marginal rate of substitution between two goods can vary though diminishing along different curves.



7. **Indifference curves for complementary goods are L-shaped**

Goods which are used jointly such as CD. and CD. Player, shirt a neck-tie, car and petrol etc. are complementary goods. So a specific combination provides satisfaction to the consumer and if one good is



increased without increasing the other one, do not provide any satisfaction to the consumer. For example, for one person only one computer is required which is point A on IC1. If we

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

increase persons without increasing computers or increase computers without increasing persons then additional computers will be useless for one person leaving his satisfaction unchanged.

BUDGET LINE OF CONSUMER

It shows the different combinations of the two goods which a consumer can purchase, while the prices of goods and income of the consumer is given. We can simply say, consumer's price line or budget line shows the purchasing power of the consumer in view of given income and prices of goods"

Mathematically it can be expressed as follows

$$I = x \cdot P_x + y \cdot P_y$$

I = Income of consumer
 X = Quantity of X purchased
 Y = Quantity of Y purchased
 P_x = Price of x
 P_y = Price of y

Suppose consumer have 100 rupees

$$I = 100 \quad P_x = 25 \quad P_y = 20$$

$$100 = 25x + 20y$$

If consumer purchases only x then putting $y = 0$

$$100 = 25x \Rightarrow x = \frac{100}{25} = 4$$

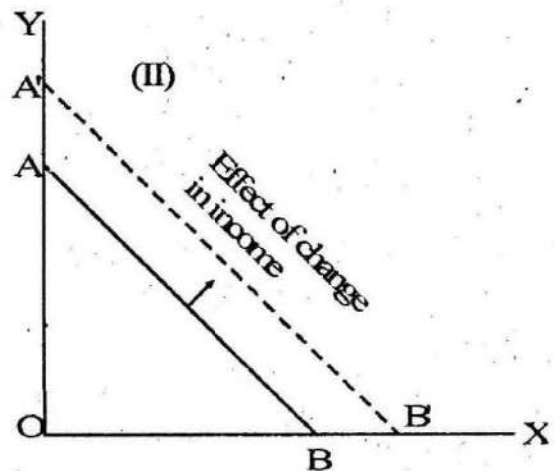
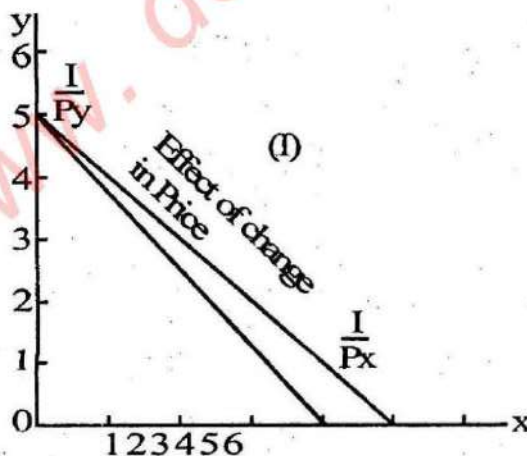
So maximum of x he can purchase is 4 units. If consumer purchases only y, putting $x = 0$ in budget equation

$$100 = 20Y \Rightarrow Y = \frac{100}{20} = 5$$

$$\text{Slope of Budget Line} = \frac{\text{Rise}}{\text{Run}} = \frac{I/P_y}{I/P_x}$$

$$\text{Solving} = \frac{I}{P_y} \times \frac{P_x}{I} = \frac{P_x}{P_y}$$

$$\text{Slope of Budget line} = \frac{P_x}{P_y}$$



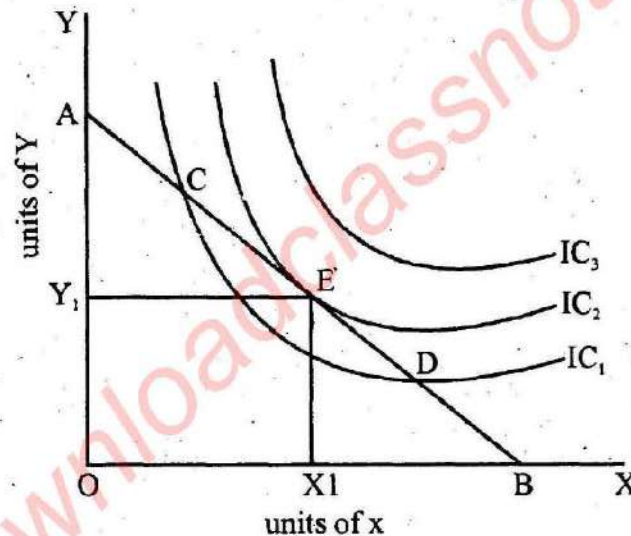
ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Now suppose if the price of one good let x falls from 25 to 20 then budget line will move outward to the right showing an increase in purchasing power of consumer in terms of x. While on y-axis it will not change. But if nominal income i.e. $I = 100$ changes then whole budget line will shift outward or inward depending upon the change. For example, if income is doubled to 200 then purchasing power of the consumer will rise in terms of both x and y and budget will move outward from both axis.

CONSUMER'S EQUILIBRIUM

Now the consumer has a map of indifference curves showing different levels of satisfaction owing to different combinations of two goods. Consumer is also aware of his purchasing power on budget line. We can find equilibrium of the consumer of these tools, i.e. maximization of his satisfaction with his limited budget.

“A consumer is said to be in Equilibrium when his budget line becomes tangent with the highest possible indifference curve”. It is explained with the help of following diagram.



In this diagram we have map of I.C. and a budget line AB. The budget line becomes tangent with the highest possible indifference curve IC_2 at point E which is the consumer's equilibrium point. At this point Consumer uses X_1 units of Commodity X and Y_1 units of commodity Y and gains maximum satisfaction

CONDITIONS OF EQUILIBRIUM

There are two conditions of equilibrium

(i) Necessary condition

Slope of I.C. = Slope of Budget line

$$\frac{M_{ux}}{M_{uy}} = \frac{P_x}{P_y}$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

(ii) Sufficient condition

I.C. must be convex to the origin at the point of equilibrium which is at point E.

So points C and D are not equilibrium points because here indifference curve is not convex to the origin. Further C & D both are on same I.C. i.e. IC_1 where as point E is on IC_2 which is higher than IC_1 . Hence E is giving more satisfaction to the consumer as compared to anyother point.

ASSUMPTIONS

Now we discuss it's assumptions.

(1) Consumer is rational

Consumer is rational and spends his income carefully to maximize his satisfaction.

(2) Utility is ordinal

Consumer can rank his preferences according to the satisfaction means he can order his preferences.

(3) Constant Prices

Consumer is well aware of the prices and these remains constant during the purchase of goods.

(4) Axiom of consistency

Consumer is consistent. If he prefers one thing on other, then he will not prefer the 2nd on first thing e.g

If $A > B$ then $B \not> A$.

(5) Axiom of Transitivity

If consumer sees that $A > B$ and also $B > C$ then definitely $A > C$

(6) Diminishing marginal rate of substitution

When good is exchanged for another good then the substitution declines because the good which consumer gives away causes his utility to rise.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

OBJECTIVE QUESTIONS

Select and tick (✓) Correct answer.

1. Human wants fulfilled free of cost are known
(a) Economic wants (b) Non-economic wants
(c) Basic necessities (d) Comforts
2. Which one is not a material good
(a) Rice (b) Book (c) Lecture of a teacher (d) Radio
3. Who discussed consumer's behaviour according to utility approach.
(a) Adam Smith (b) J.R. Hicks
(c) Prof. Robbins (d) Alfred Marshall
4. Keeping in view of following data, answer the following questions.

Units of good	Total utility
1	15
2	25
3	30
4	25
5	23

(a) Find out point of satiety.
(b) At which unit M_u is negative
(c) Find out Marginal utility (M_u) at 2nd unit.
5. Which law will be applicable when Marginal utility of the good under use goes on diminishing?
(a) Law of diminishing marginal utility (b) Law of decreasing return
(c) Law of decreasing cost (d) Law of indifference
6. Which law operates when M_u of all the purchased goods becomes equal.
(a) Law of demand (b) Law of diminishing marginal utility
(c) Law of Equi-Marginal Utility (d) Law of constant Return
7. When different combinations of two goods provide equal level of satisfaction to the consumer. He acts upon the theory of?
(a) Law of diminishing marginal utility (b) Indifference curves approach.
(c) Law of increasing return (d) Law of increasing cost.
8. Which one is not a capital good as under:-
(a) An iron for commercial use (b) Sewing Machine of a Tailor
(c) Tractor of a cultivator (d) Wall clock in the house.
9. What is the basic preference for human wants
(a) Non-economic wants (b) Basic necessities
(c) Luxuries (d) Non material wants
10. What is the special feature of an indifference curve.
(a) Vertical (b) Horizontal
(c) Convex to origin (d) Concave from origin.
11. The goods which provide direct satisfaction to human wants are known
(a) Capital goods (b) Consumer's goods
(c) Free goods (d) Exports

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

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12. According to Modern economists utility is concerned with consumer's aspect:-
(a) Material (b) Social (c) Psychological (d) Ethical
13. When $Mu=0$, Total utility is:
(a) Maximum (b) Minimum (c) Falling (d) Rising
14. What is the slope of Mu curve.
(a) Positive (b) Horizontal (c) vertical (d) Negative
15. Who introduced the first law of consumption?
(a) Adam Smith (b) Prof. Robbins
(c) Alfred Marshall (d) Robinson
16. Rate of change of the units of two goods is known
(a) Discount rate (b) Marginal rate of substitution (MRS)
(c) Marginal efficiency of capital (d) Foreign exchange rate
17. Two indifference curves do not.... Each other.
(a) Horizontal (b) Follow (c) Intersect (d) Compete
18. Law of equi-Marginal utility is also known:-
(a) Law of diminishing marginal utility (b) Law of extreme satisfaction
(c) Law of indifference curves (d) Law of ordinal measurement of utility
19. Mother's love and friendship of a friend are related with:
(a) Economic wants (b) Material wants
(c) Non-economic wants (d) Rare wants
20. Which law provides basis for the law of demand.
(a) Law of diminishing marginal utility (b) Law of substitution
(c) Law of supply (d) Law of ethics

DESCRIPTIVE PART - I

SHORT NOTES

1. What is meant by Economic Wants?

The wants for which we have to pay some price or have to participate in economic activity such as food, clothing housing etc. are regarded as economic wants. These may be satisfied by economic goods/Services. Rendered by Doctors, Professors, Banks and other institutions.

2. What are non-economic wants?

The wants which are fulfilled without any effort or without indulging in any economic activity are called non-economic wants, Such as oxygen, sunlight rain, cool breeze, mother's affection etc. These are also regarded as free wants.

3. Bring out difference between material and non-material goods?

The goods which are tangible, having weight and volume such as wheat, cloth, car, watch etc. are called material goods, The goods which neither have volume nor are tangible are

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

called non-material goods or services such as doctor examines patient, professor teaches and lawyers services etc.

4. What are consumption goods?

The goods which directly satisfy human wants are known as consumption goods. For example, water directly quenches human thirst, food satisfies hunger, Services are also included in consumption goods because they also directly satisfy human wants.

5. Define Capital goods?

Some goods do not directly satisfy human wants but these goods are used to produce certain other goods which satisfy human wants such as we demand bricks, sand, cement, iron, wood labor etc. to fulfill our desire of building a house which is a "Consumer good".

6. Point out any five Characteristics of economic wants.

- 1) Human wants are unlimited.
- 2) Human wants are recurring.
- 3) Differ in importance
- 4) Alternative methods of satisfaction
- 5) Complementary wants.

7. Human wants can be divided into how many parts?

Human wants can be divided into two parts i.e. Economic wants and non-economic wants.

8. What is utility?

The quality of a good which satisfies certain human wants is called utility. For example, Water has the utility to quench thirst, bus has utility for transportation, pen has utility to write, watch tell us time etc.

9. Bring out the differences between utility and usefulness?

The goods which may have utility for a person not necessarily would be beneficial for him such as wine and drugs may have utility for a user's of these items, but these are not good for their health. These are harmful for human beings physically as well as morally.

10. Why indifference curves are negatively sloped?

Indifference curves are negatively sloped due to diminishing marginal rate of substitution (DMRS). Because in order to keep same level of satisfaction consumer gets more of one good by decreasing the units of the other good.

11. What is the relationship between (TU) Total Utility and (MU) Marginal Utility.

- With the continuous use of units of goods TU rises and MU remains positive.
- When TU become maximum, $MU=0$
- When TU Starts falling, MU Becomes negative.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

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12. Define law of diminishing Marginal Utility

According to DR Alfred Marshall

“Other things remaining the same, marginal utility becomes smaller and smaller, the more of a good the person has”. With the use of an additional unit of a good marginal utility goes on diminishing.

13. Point out four assumptions of law of diminishing marginal utility?

Continuous use of the good.

- Reasonable quantity (neither too big nor too small)
- Homogenous nature of good. (All units are equal in every respect)
- State of mind of the consumer remains the same.

14. Write down “three exceptions” of the law of diminishing marginal utility?

The things upon which this law does not apply are two on as Its exceptions: such as;

- (i) Wealth and money.
- (ii) Knowledge and fine Art.
- (iii) Rare and most precious goods.

15. Bring out “Four limitations of the law of equi-marginal utility.

- (i) Cordinal measurement of utility is not possible.
- (ii) No rational and conscious calculations.
- (iii) Ignorance of consumer.
- (iv) Indivisibility of certain goods such as; a cow, sewing machine, wrist watch etc.;

16. Define law of “Equi – Marginal Utility”?

It can be defined as;

“A consumer is said to be in equilibrium when he spends his money on various goods in such a way that MU of all goods equal to each other and MU of every rupee spent on

each good is equal” $\frac{M_u_a}{P_a} = \frac{M_u_b}{P_b}$

17. What is the difference between “Comforts and necessities”?

The goods which are necessary for the survival of human beings such as food, clothing, shelter, medicines etc. are called necessities, whereas the goods which facilitate human efforts in achieving satisfaction such as motorcycle watch, radio, electric fan etc. are called comforts.

18. Discuss various aspects of utility?

- “Initial utility” Which is achieved by using first unit of good.
- “Marginal Utility” it is achieved by an additional unit of the good.
- “Positive Utility” When TU keeps on increasing and MU has not become Zero.
- “Zero Utility” At point of satiety TU becomes maximum and MU become Zero.
- “Negative Utility” When TU starts falling and MU becomes negative (after being zero)

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

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19. Define an Indifference Curve?

“An indifference curve is a locus of points which yield same level of satisfaction to which the consumer remains indifferent” Any combination on an indifference curve gives equal level of Satisfaction.

20. Bring out four characteristics of Indifference Curves?

An IC slopes downward from left to right.

- IC are convex to the origin.
- Never touches any axis
- Never intersect each other.

PART – II

1. Discuss salient features of human wants.
2. Explain graphically the relationship between Total Utility and Marginal Utility.
3. Explain “law of diminishing marginal utility” with the help of schedule and diagram.
4. Explain “Consumers Equilibrium” with respect to “Utility approach”.
5. What is an indifference curve. Bring out salient features of indifference curves.
6. Explain “consumer equilibrium” with the help of indifference curves.

